

## (a) Rejecting Invalid Questions

### R-Zero

Questioner



No real solutions?  
 But I have to answer...



### Unstable Answer

Rollout 1:  
 $x^2+y^2=3$

Rollout 2:  
 No real solution

Rollout 3:  
 $x^2+y^2=3$

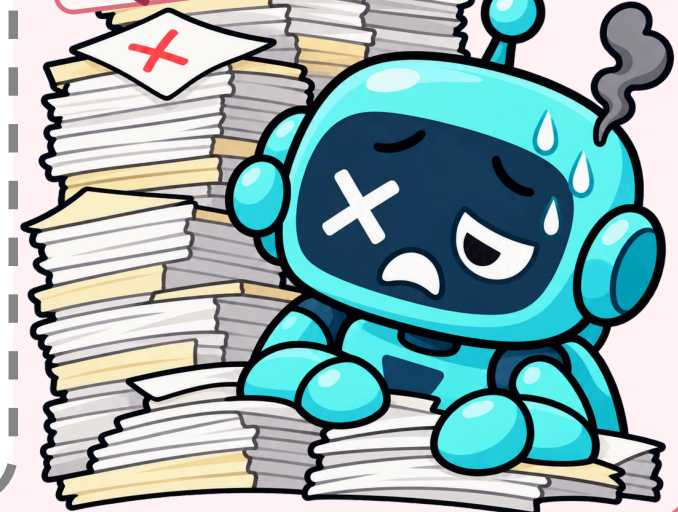
...

High Uncertainty  
 Treated as Hard



Select for  
 Training

Invalid  
 Questions

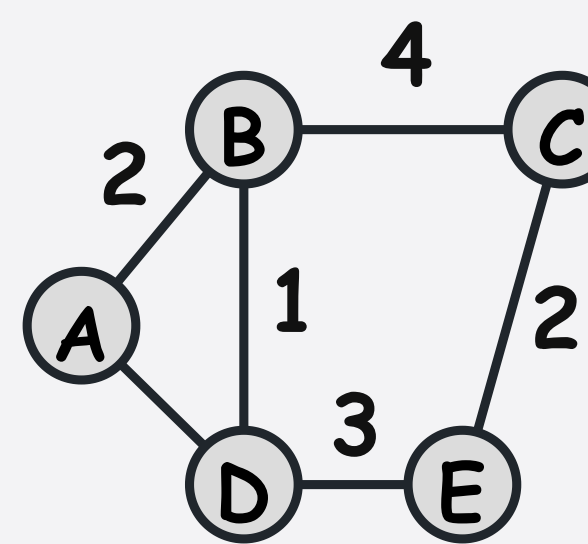


## (b) Recognizing Repeated Tasks

Same Problem,  
 Different Costume

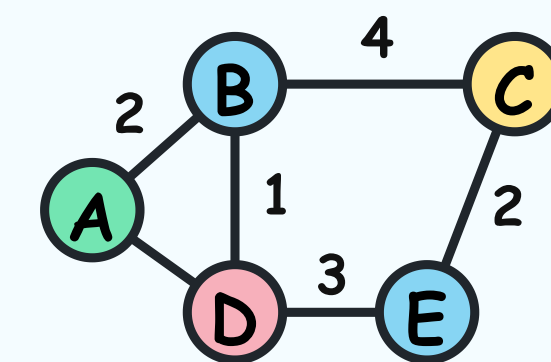


Same  
 Underlying  
 Task



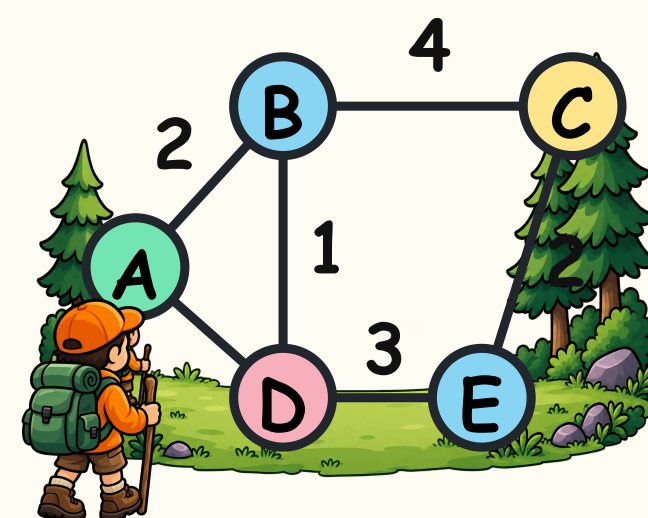
### Version A

Find the shortest  
 path from A to E.



### Version B

A backpacker crosses  
 a magic forest. Which  
 route uses the least  
 energy?



### R-Quest



INVALID  
 Won't Answer

After Validity Initialization  
 Detects Invalid Question

$(x-y)^2 = (x+y)^2 - 4xy$   
 $= 9-12 = -3 < 0$   
 No Real Solution!



Rejected &  
 Filtered Out



Negative  
 Reward



Only Valid  
 Questions Earn  
 Rewards!

### R-Zero

Same task,  
 over and over...



Version A

Version B

Low Lexical  
 Similarity

Weak Repetition  
 Penalty  
 Treated as Novel



Select for  
 Training

Different  
 Words, Less  
 Penalty!

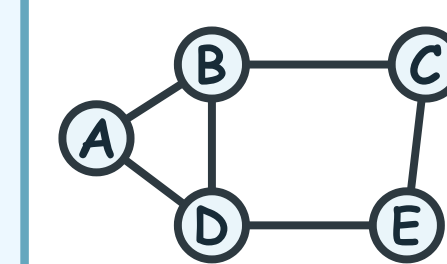


### R-Quest

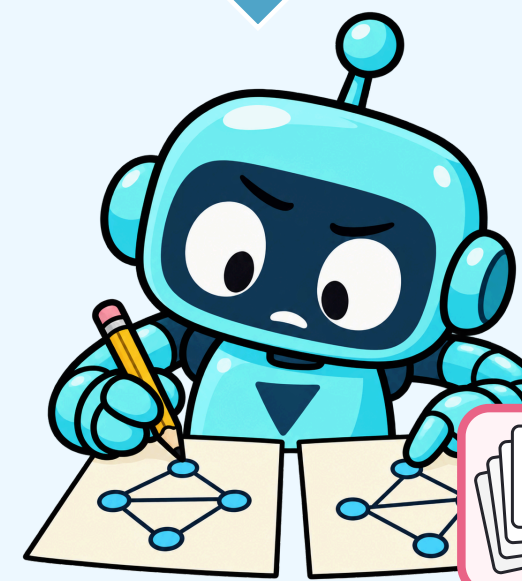
Version A

Version B

Same Task:  
 Shortest Path



Frequent in  
 Training Data



Rejected &  
 Filtered Out

Only Novel  
 Questions Earn  
 Rewards!

